Computational Evidence: Monodromy of Inverse z^z

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1 Setup

Solving $z^z = w$ requires choosing a logarithm branch $k \in \mathbb{Z}$ and a Lambert-W branch $m \in \mathbb{Z}$. We test whether these two branch parameters behave independently under monodromy by tracking the inverse along closed loops in the w-plane.

Method. Nearest-neighbour continuation: at each step of a discretized loop $\gamma(t_j)$, enumerate candidate preimages from nearby (k, m) branches and select the closest to $z(t_{j-1})$. This is heuristic, not rigorous analytic continuation.

Lift coordinates. We also track input-derived coordinates $k_{\text{lift}} = \text{unwrap } \arg(\gamma)/(2\pi)$ and $m_{\text{lift}} = \text{unwrap } \arg(\log \gamma + 1/e)/(2\pi)$. **Caveat:** these are functions of the input loop, not the tracked inverse; their continuity is partly by construction. The substantive evidence comes from the gap and sheet-shift tests below.

2 Experiments

Two loops: Loop A around $w^* = e^{-1/e}$ (radius 0.40), Loop B around 0 (radius 2.00).

2.1 Monodromy gaps (Scripts 01–02)

	Gap $\lvert z(2\pi) - z(0)\rvert$	Spike ratio
Loop A (w^*)	1.096	2.08×
Loop B (0)	3.147	1.37×

Table 1: Both loops produce nonzero closure gaps in $\mathbb{C}$. Low spike ratios show the gap accumulates smoothly, not from a single numerical discontinuity.

2.2 Branch independence (Scripts 02–03)

2.3 Closure modulo sheet shift (Script 03)

Endpoints match after applying the tracked integer shift: $z(2\pi) \approx T_{(\Delta k, \Delta m)}(z(0))$ with residuals $\sim 10^{-17}$ (machine epsilon). The path is open in $\mathbb{C}$ but closed on the covering space.

	Integer branch shift		Lift-coord. displacement	
	Δk	Δm	Δk_{lift}	Δm_{lift}
Loop A	0	1	≈ 0	1.000
Loop B	1	0	1.000	0.223

Table 2: Integer shifts are cleanly independent: $(0, 1)$ vs $(1, 0)$, residuals at machine epsilon. The 0.223 in Δm_{lift} for Loop B reflects input-loop geometry (Loop B passes near $-1/e$), not a branch coupling — the integer Δm is exactly 0.

n	Gap A	Resid. A	Gap B	Resid. B
100	1.096153	2×10^{-17}	3.147208	0
1600	1.096153	2×10^{-17}	3.147208	0
6400	1.096153	2×10^{-17}	3.147208	0

Table 3: Gaps and shifts are identical from $n = 100$ to $n = 6400$.

2.4 Resolution stability (Script 03)

3 Figures

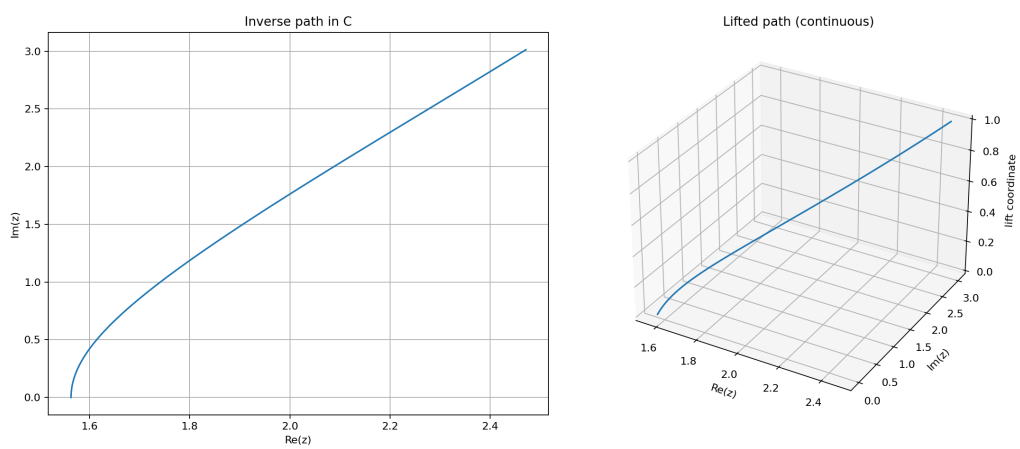

Figure 1: Experiment 1. Tracked inverse in $\mathbb{C}$ (left) and lifted view (right).

4 What This Does and Does Not Show

Shown:

- Nontrivial monodromy: closed loops in w produce open paths in $\mathbb{C}$.
- Two independent integer branch generators: shifts $(0, 1)$ and $(1, 0)$.
- Deck-action closure with machine-epsilon residuals.
- Perfect stability across discretisation resolution.

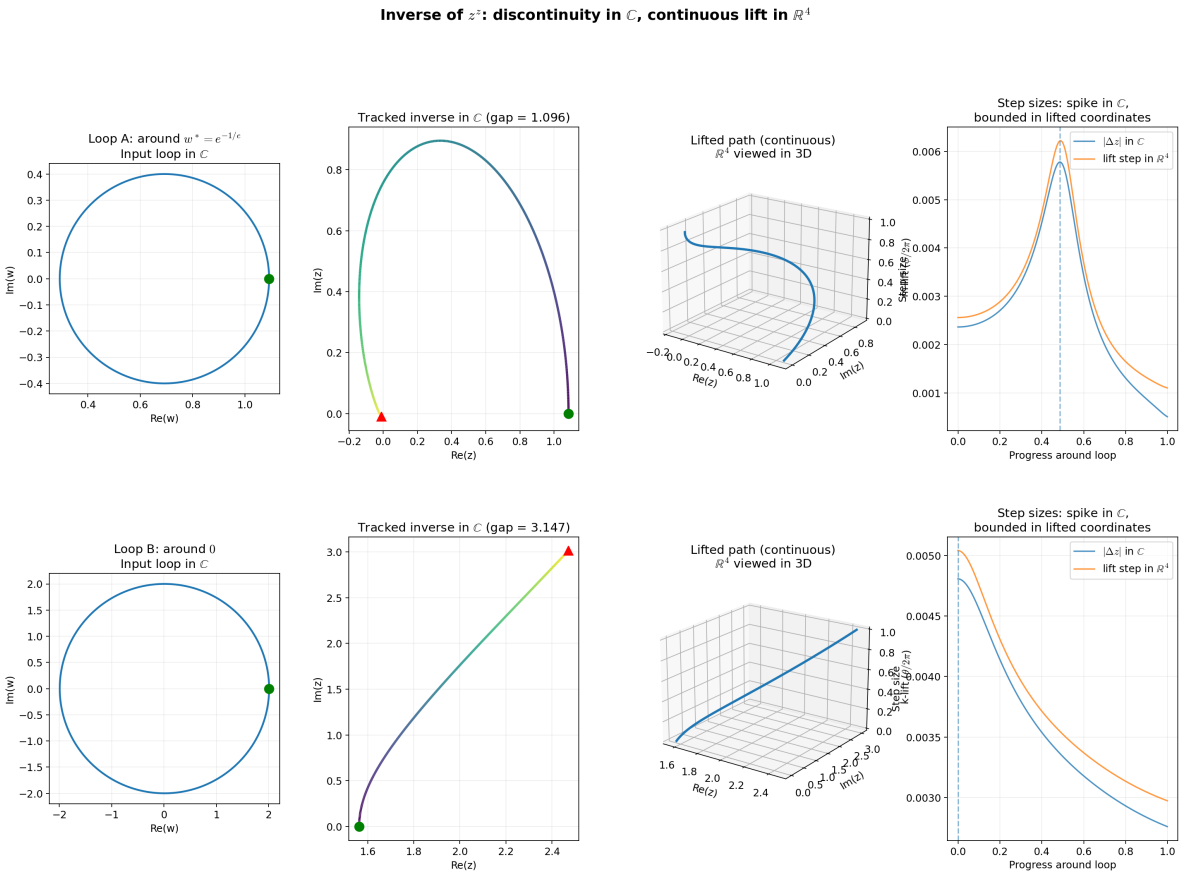

Figure 2: Experiment 2. Two-loop comparison with gaps and lift diagnostics.

Not shown:

- That monodromy of z^z is multivalued is already known; these experiments demonstrate it, they do not discover it.
- The experiments show $\mathbb{C}$ is insufficient, not that 4 dimensions are the *minimum*. A lower bound needs a covering-space argument.
- Only two loops are tested. The monodromy group could differ from $\mathbb{Z} \times \mathbb{Z}$; compositions and higher-winding tests are needed.
- Nothing here tests quaternionic structure. The $\mathbb{H}$ aspect of the conjecture has zero computational support.
- Nearest-neighbour continuation is heuristic (search radius $|dk|, |dm| \leq 2$, no formal error bound).

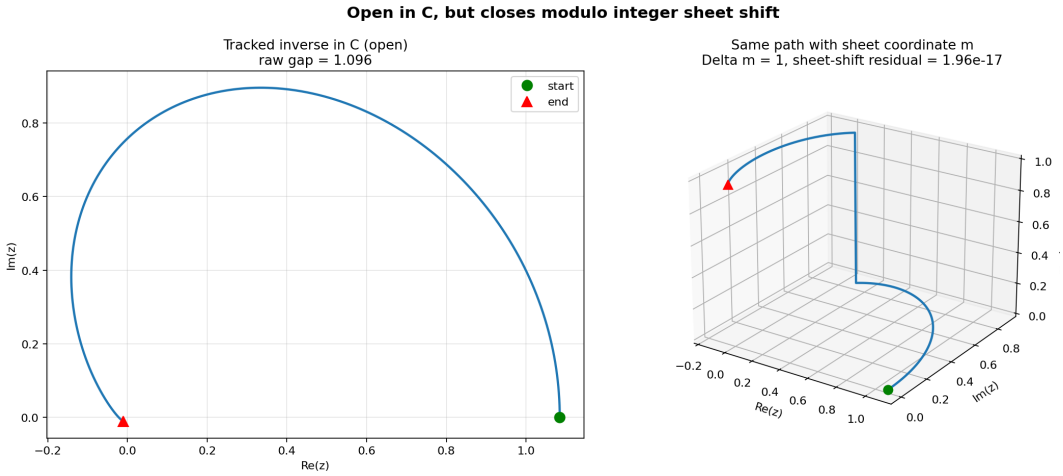

Figure 3: Experiment 3. Open in $\mathbb{C}$, closed modulo sheet shift.

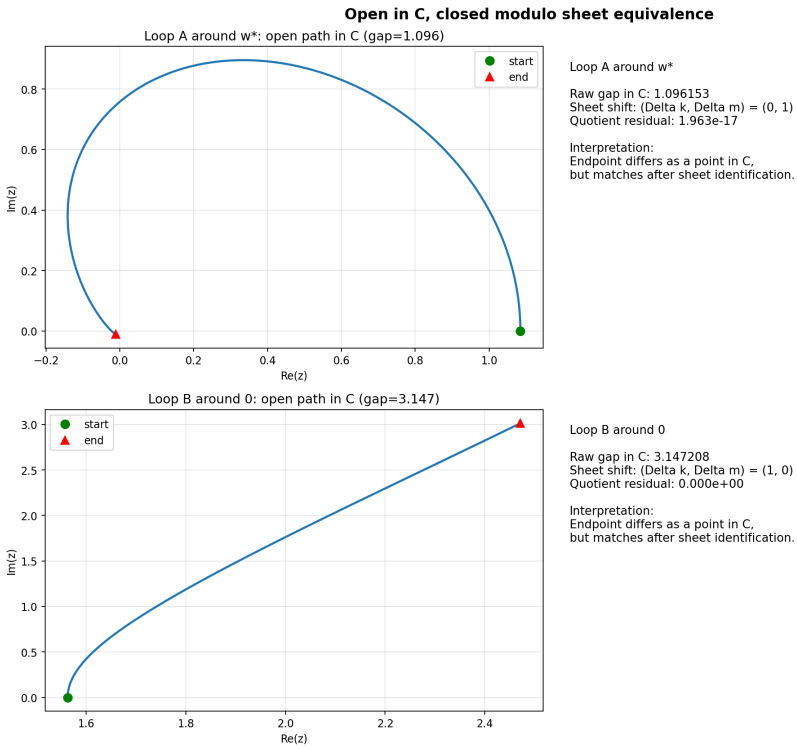

Figure 4: Experiment 4. Summary: quotient-closure with metrics.